Single-View Human Reconstruction from Egocentric Fisheye Cameras

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Abstract

Egocentric Fisheye Cameras have found wide appeal in xR research for pose estimation due to their attachment to head-mounted devices and wide field of view, overcoming spacial constraints faced by outside-in tracking.
In a related area, recent research has explored to generate reconstructions of such images, but the limited view of the body poses a challenge in the retrieval of sufficient context for an accurate representation. Therefore, many procedures still rely on data recorded from additional perspectives.
This report documents a method, which attempts to merge advancements in fisheye pose estimation and generative image models in order to produce a fully textured, posed SMPL mesh from on a single monocular RGB Fisheye image. However, in evaluation the procedure shows suboptimal performance, failing to match available information with the posed mesh across all datasets.

Introduction

Pose estimation is the task of determining the positions of an individual's joints from input image or video data obtained from one or multiple perspectives. Historically, pose estimation relied on the use of physical markers applied to the persons body. However, recent advancements have enabled markerless tracking, even with a limited number of camera views. Furthermore, significant progress has been made in Inside-Out tracking, which employs fisheye cameras mounted on the subject's head as an alternative to traditional Outside-In tracking. This approach offers several advantages, including reduced susceptibility to occlusion, a common issue in Outside-In tracking, and compatibility with environments of limited space, where fixed camera placement is not possible. Additionally, the large point of view provided by fisheye lenses allows high visibility of most limbs across a broad range of poses. Earlier methods relied on stereo setups for egocentric pose estimation to infer depth data [8]. However, more recent approaches achieve greater accuracy using a single monocular fisheye lens without requiring additional context [10].

Building upon existing methodologies in 3D pose estimation, various approaches have also been proposed for reconstructing a subject's appearance using visual image or video data in combination with a pose prior as an input. One goal of 3D Human Reconstruction is to generate a realistic representation of the subject, enabling their appearance to be rearranged into novel poses based on the underlying skeletal structure. In this area of research, many existing methods have demonstrated impressive results in Outside-In reconstruction. However, reconstruction from an Inside-Out fisheye perspective presents greater challenges, as this viewpoint captures the body from the front in a highly distorted and foreshortened manner, providing only limited information within a single frame. To address these constraints, current approaches in Inside-Out reconstruction condition their estimations on previously computed or live Outside-In views of the subject, thus mitigating the limitations imposed by the restricted field of view [2].

A successful merge of both techniques, would allow for real world practical use in xR Applications. For example, it could enable the generation of an accurate digital avatar based on the user's real appearance without necessitating any additional setup. This could be achieved solely through the utilization of an egocentric lens embedded in the user's headset.

This report presents a procedure that integrates contemporary methods to develop a unified model for Inside-Out egocentric pose estimation and appearance reconstruction. The proposed model first estimates pose using a single monocular fisheye image captured from an Inside-Out perspective, leveraging an existing pose estimation network. Next, it maps the available textural information onto an SMPL mesh posed according to the estimated pose. Finally, the UV map, containing the mapped texture, is passed to a generative model, which inpaints the remaining areas based on the available data. This process results in a

fully textured, posed SMPL mesh, where missing texture information is enhanced through generative models. The final mesh can be viewed from any angle and reposed into different positions.

Related Works

3D Human Reconstriction

The reconstruction of human subjects from monocular images has been explored by various researchers over the years, employing a variety of approaches and methodologies. One notable approach are reconstructions based on combining classical skeleton prediction techniques with modern splatting-based rendering approaches. Many methods also utilize the SMPL template as a foundational mesh to project and map textural data onto, a procedure that likewise inspired the methodology of this work.

The authors of the paper "Gaussian Articulated Template Models" (GART) [3] followed such an approach by first training a Neural Radiance Field (NeRF) [5] given a monocular video of a human subject as an input and approximate the subject's appearance in canonical space. The canonical appearance is represented explicitly as a set of Gaussians. This Gaussian-based representation is subsequently skinned to a Skinned Multi-Person Linear model (SMPL) [4] skeleton template, enabling free articulation through the adjustment of SMPL pose parameters. The articulated Gaussian representation is governed by a set of learnable skinning weights, which dictate the rate of deformation in accordance with the skeletons movement. The freely articulated representation can then be rendered in real time using Gaussian splatting.

Another method build on on a similar approach is "Holistic Urban 3D Scene Understanding via Gaussian Splatting" (HUGS) [13]. The method begins by performing a reverse estimation of SMPL pose and body shape parameters from a set of monocular image or video data. Given that estimation of the body shape, a SMPL skeleton is positioned in a T-Pose within canonical space. Subsequently, a MLP trained on the video data is employed to populate the canonical space with Gaussians. The MLP takes spatial coordinates as input and predicts the parameters of the Gaussians, including RGB values, opacity, spherical harmonics coefficients, and linear blend skinning (LBS) weights. Similar to the GART approach, the resulting skinned model can be articulated and rendered efficiently in real time through Gaussian splatting.

Fisheye Pose Estimation

The estimation of human poses using fisheye imagery has been the focus of various studies in recent years, leveraging unique characteristics of fisheye cameras such as their wide field of view and ability to capture extensive spatial information.

A notable approach is presented in xR-EgoCap [8], which employs a stereo fisheye camera setup for pose estimation. The method initializes by determining bone length parameters from external views before estimating pose through an analysis-by-synthesis approach. A CNN-based limb detector predicts bone center positions, generating heatmaps that are approximated with colored Gaussians. The framework also optimization terms to enforce pose rationality based on temporal pose difference, refining the final estimation.

A related approach is proposed in the xR-EgoPose paper [10], which introduces a method for pose extraction based solely on monocular image data from a synthetic, labeled fisheye pose dataset developed by the authors. The approach follows a two-step process: first, a modified ResNet extracts probability heatmaps for each limb, which are then processed by a 3D pose estimator. This network computes an encoding of the heatmap and branches into two paths, one predicting 3D limb positions and the other reconstructing the heatmap to enforce consistency. The method optimizes pose accuracy using a loss function that incorporates ground-truth 2D and 3D pose supervision, along with a reconstruction loss between the original and decoded heatmaps. According to the authors, this dual-path structure improves prediction stability and helps the model learn a better probability distribution of the input heatmap.

Egocentric pose and appearance estimation

A method similar to the goal of this project was recently published by the authors of EgoAvatar [2]. In their work, the authors propose a pipeline that enables real-time tracking of human pose and the mapping of a Gaussian representation of visuals onto the predicted pose. In addition to rigid deformations based on the underlying skeleton, the approach incorporates dynamic clothing deformations that respond to temporal movement. The pipeline processes monocular egocentric fisheye images, extracting joint keypoints to predict a skeleton via an inverse kinematic solver. A mesh is then fitted to the predicted skeleton and a synthetic egocentric view of the subject is generated. This synthetic view is compared against the ground-truth silhouette using an MLP, enabling further refinement of the predicted mesh. The texture is generated by projecting Gaussians onto the mesh's UV space via a neural

network, which are subsequently rendered back into 3D space to produce the final pose. While this process works during inference, the model requires pretraining on the individual's shape and visual data. The skeleton prediction, mesh extraction, and Gaussian splatting steps necessitate pretraining based on multi-view outside-in video tailored to each specific subject.

Methods

This project proposes a method to address two key challenges in egocentric imaging and rendering. The first challenge is the estimation of a subject's pose from a single monocular egocentric fisheye image. The second challenge involves the extraction of available appearance information from the same image. Both types of information are subsequently combined onto a single mesh, with missing appearance data inpainted based on the existing information. To tackle these challenges, publicly available models and methods are systematically applied, with data adjustments and mapping procedures integrated at intermediate stages to generate the final mesh.

Pose Estimation

For the purpose of pose reconstruction, the SceneEgo model [12] was selected. The reasoning behind this choice was the availability of its source code and trained model weights, as well as its relatively strong performance compared to similar methods, such as xR-EgoPose [10], for which it demonstrates superior prediction accuracy according to the authors' publication on the method. Although the author has released a more recent procedure with improved results using EgoWholeMoCap [11], the SceneEgo model was preferred due to its stepwise pipeline approach, which produced artifacts at each stage that could be leveraged in subsequent steps of the procedure.

SceneEgo's pose reconstruction pipeline can be categorized into three primary steps to produce an accurate skeleton in a specific pose, given a frame of an egocentric image as input. These steps are: Image Segmentation, Depth Estimation, and Pose Estimation.

The first step of the pipeline employs a segmentation model that processes the egocentric image frame I to produce a probability mask S^P, indicating the likelihood of each pixel belonging to a human. Applying a threshold filter converts this into a binary mask S.

Next, the binary mask S and image I are fed into the depth prediction model, which generates a depth image D^B. The mask is used to remove depth values corresponding to the subject, yielding $D^M = (1 - S) \odot D^B$. Missing depth data is then filled by a depth prediction network, resulting in the final scene depth image D^S. Simultaneously, a body joint prediction network generates a 2D pose heatmap $F^{H \times W \times 15}$ for 15 joints from the input image I.

The scene depth map D^S and pose predictions F are projected onto a uniform 3D voxel space $V^{L \times L \times L}$ centered within a $2m$ area beneath the egocentric camera. The depth map is reverse projected from fisheye image coordinates into canonical space, forming a point cloud that is sampled into the voxel grid V^S. Similarly, body pose heatmaps are reverse projected into the voxel space V^B. Both V^S and V^B are processed by a Voxel2Voxel network, producing 3D heatmaps V_{heatmap} for each joint, which are subsequently regressed into the final 3D skeleton prediction. The authors highlight that including the scene's voxel representation alongside the initial pose improves prediction plausibility by providing context on the scene's height and possible body contact points. For further details, refer to the original paper [12].

The resulting skeleton takes a simplified shape and is described by a total of 15 joints modeling each part of the body, excluding the head. In addition, the joint predictions are not limited by realistic constraints such as bone lengths or body symmetry to a stricter extend than is naturally trained by the models heat maps. Therefore inconsistencies can occur, like asymmetry in bone length or unrealistic proportions of appendages, which have to be accounted for in further processing.

To make the joint estimation more streamlined and allow compatibility with contemporary visual methods, the simplified estimation of 15 joints has been used to derive a SMPL-Skeleton representation of the same pose. For this purpose, the simplified pose estimation has been extended to 24 key points by interpolating the missing joints, and overlaid over the SMPL Skeleton at the pelvis joint. Subsequently, correspondences between the joints of the simplified source skeleton and the 24-joint SMPL target skeleton are established though means of gradient descent optimization to minimize the L2 norm:

$$E = ||J_s - J_T||_2 + w \cdot ||\theta_T - \theta_{\text{Rest}}||_2 \quad (1)$$

where $J_S \in \mathbb{R}^{3 \times N}$ and $J_T \in \mathbb{R}^{3 \times N}$ represent the joint positions of the source and target skeletons, respectively, with column-matched correspondences. The second term, weighted by w, penalizes deviations of the target pose parameters θ_T from the initial "T-Pose" resting pose parameters θ_{Rest}, ensuring stability by preventing unreal-

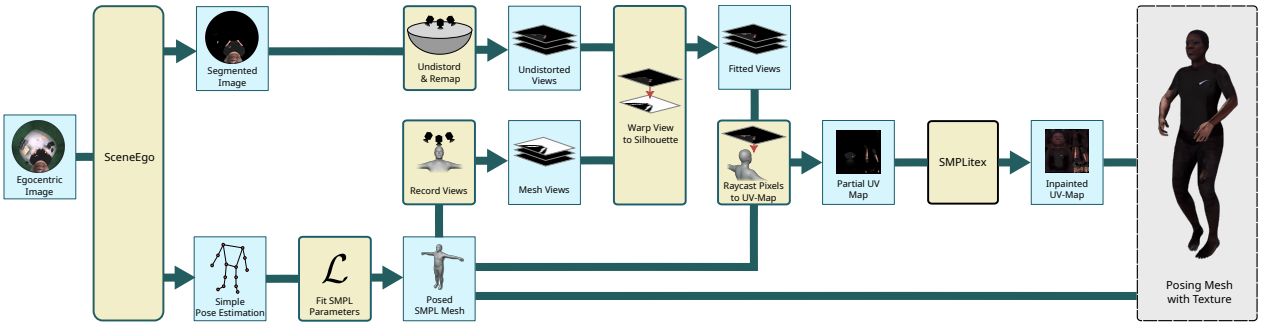


Figure 1. **High level overview** of the Procedure's pipeline. The procedure follows a pipeline approach, combining partial results across multiple stages to generate the final SMPL UV map and pose parameters from a monocular egocentric fisheye image input. Methods are denoted by yellow backgrounds, while blue backgrounds describe artifacts obtained as partial results by the pipeline.

istic joint rotations or deformations due to differences in joint proportions. It is also to note, that the rotation of joints is not modeled, as it is not predicted by SceneEgo. Therefore for each Position a resting rotation is assumed. Additionally, the body shape parameters β of the SMPL object are kept unchanged to their initialized values.

Fisheye Undistortion

For the generation of pose visualizations, the highly distorted fisheye images were undistorted to resemble images captured under the pinhole camera assumption. This approach was chosen to facilitate subsequent mapping and warping procedures and allow for compatibility with available methods. The undistortion process consisted of estimating the intrinsic camera matrix K_F of the fisheye camera using a set of checkerboard patterns and the OpenCV library. Based on K_F and the associated distortion coefficients, multiple undistortion parameters and their corresponding camera matrices were computed. This enabled the re-rendering of image portions at various angles, yielding undistorted and rectified views under the pinhole camera assumption.

Pose Visualization

As images captured by an egocentric fisheye lens show only a limited portion of the front body and are heavily foreshortened, extracting a person's appearance poses a significant challenge without external views. To transfer the person's appearance, the limited image data is used to condition a generative image network that inpaints occluded or missing body regions. For this purpose, the SMPLitex diffusion model, a generative model designed to generate SMPL UV maps, is employed [1]. The model is capable of inpainting UV maps based on partial input, enabling texture completion. To apply this process, the texture from the original image I is projected onto the SMPL mesh using the estimated body pose.

For this purpose, first, an SMPL mesh $\mathcal{M}(\theta_T)$ based on the fitted pose parameters θ_T is generated by utilization of the Blender SMPL-X add-on [7]. As opposed to the SMPLX Python library, this add-on allows the export of the SMPL Mesh, while keeping the correspondence between faces and the UV-Texture intact. For completeness, it is important to note that, while the add-on operates on the extended SMPL-X format, the exported mesh still supports the legacy SMPL UV-Mappping, which is accepted by the SMPLitex model.

In order to match visuals to the generated mesh $\mathcal{M}(\theta_T)$, the image I is first masked by the binary mask S in order to filter out the surroundings from the image. This segmentation is provided as an artifact by SceneEgos depth map calculation. It is then undistorted based on the undistortion parameters computed in the first section, which results in a set of 3 images I_{U1}, I_{U2} and I_{U3}, which describe a portion of the Fisheye image undistorted from their corresponding rotation matrices R_{U1}, R_{U2}, R_{U3}, each of them displaying a partial view of the body (See Figure 6).

To achieve an equivalent rendering of the mesh for texture transfer, the camera extrinsic and intrinsic parameters must be estimated for the mesh. The extrinsic parameters E_{U1}, E_{U2}, and E_{U3} are defined relative to the pelvis joint position, considering the camera's assumed location relative to this reference point. The corresponding image rotation R_{U1}, R_{U2} and R_{U3} is then applied to align the mesh view with the image viewpoint. The intrinsic parameters are represented by the camera matrix calculated in the previous section, assuming the same camera model as used for the undistorted images.

Given the Camera parameters, the generated mesh $\mathcal{M}(\theta_T)$ can now be rendered from the same view, as

depicted in the undistorted images I_{U1}, I_{U2}, I_{U3}. Since the body parameters of the model remain unadjusted and the pose prediction does not fully align with the mesh, additional preprocessing is performed to achieve an exact mapping between image and render. To address this, the silhouette of the mesh is rendered onto images S_{U1}, S_{U2}, S_{U3} for each view. Then the images I_{Ui} are stretched and interpolated to match the dimensions of non zero pixels in height of their silhouette counterpart S_{Ui}. This is followed up by a row-wise mapping where each non-zero pixel w_I in a row u of image I_{Ui} is mapped to non zero pixels w_S in the same row of image S_{Ui}. Depending on the difference in non-zero pixels in a given row, some values in I_{Ui} are stretched or compressed to fit the silhouette of S_{Ui}. A visual representation can be seen in figure 6.

Having established a 2D correspondence, this property is extended to 3D by utilizing the camera parameters to raycast each pixel into 3D space, determining intersections with the mesh's triangles. The corresponding location on the UV map is then deduced via barycentric interpolation, where the pixel value is applied. Due to foreshortening effects, the pixel density of mapped regions decreases with increasing distance from the camera. To mitigate this, smaller segments are dilated to cover a larger area on the UV map, maximizing the context provided to the generative model.

The preprocessed UV map serves as input to the generative model, which completes the remaining untextured regions of the mesh. By subsequently mapping the completed UV map back onto the SMPL mesh, the final output, a fully textured 3D mesh of the predicted pose, is obtained, enabling visualization from novel viewpoints.

Evaluation

Datasets and Parameters

Pose estimation and visual reconstruction were evaluated on three datasets: the synthetic EgoGTA dataset [12], the real-world SceneEgo test dataset, and a small self-recorded dataset. The synthetic dataset features egocentric views of virtual characters in different clothing, set against 3D scene backdrops depicting characters either walking or sitting upright. The SceneEgo dataset contains real-world recordings of people performing a variety of poses and actions. Finally, the self-recorded dataset captures various poses, some of which are not represented in the SceneEgo dataset, under more challenging lighting conditions. A sample of the datasets can be seen in the figures 3, 2 and 4.

For visual reconstruction, the SMPLX Neutral 1.1 mesh [7] served as the base with unmodified body parameters, ensuring compatibility with SMPLitex through the SMPL 2021 UV mapping. Pose fitting applied a weight parameter of $w = 0.1$ to moderately penalize deviations from the SMPLX rest position across 200 iterations. For Texture-Mesh mappings, 3 different views were used with 45 angles between each view. Recordings made to determine camera matrices and undistortion parameters were shot using a FLIR Chameleon camera with a fisheye lens attachment.



Figure 2. A sample of meshes generated from the SceneEgo dataset. The bottom left of each sample displays the original fisheye view, the top right shows the inpainted UV map, and the right side presents the complete mesh from both front and back views.

Measures

Since the SceneEgo model was used without modification and was trained on both EgoGTA and the self-recorded dataset, which lacks labeled ground truth 3D joint locations, no quantitative evaluation of pose accuracy was conducted, as it would be redundant to the results presented in [12]. Instead, the evaluation focused solely on the visual quality of the reconstructed model and the impact of the pose estimation fitting process on the final output. For quantitative metrics, I refer to the original evaluation by the authors [12], which remains fully representative to the predicted joint position estimations in this procedure.

Results

The proposed method exhibits generally suboptimal performance across all datasets, achieving moderate accuracy in transferring static body regions, such as the torso, onto the UV map, while struggling with dynamic regions, particularly the limbs and shoulders.



Figure 3. A sample of meshes generated from the EgoGTA dataset. The bottom left of each sample displays the original fisheye view, the top right shows the inpainted UV map, and the right side presents the complete mesh from both front and back views.



Figure 4. A sample of meshes generated from the Custom dataset. The bottom left of each sample displays the original fisheye view, the top right shows the inpainted UV map, and the right side presents the complete mesh from both front and back views.

Skeleton fitting

In most cases, the skeleton fitting to the pose yields generally good results, often closely matching the predicted 15-joint pose while maintaining plausible joint rotations. However, outliers occur in images where SceneEgo's pose prediction produces highly inaccurate results due to lower body occlusion. This is evident in Figure 4 (top right) and Figure 2 (top left), where the leg position and rotation parameters assume highly unrealistic values. In general observation, the upper body of skeletons seem to match the ground truth to a higher degree than the lower body due to the SceneEgo struggling to produce leg estimations for the EgoGTA and Custom dataset, as seen in Figure 5.

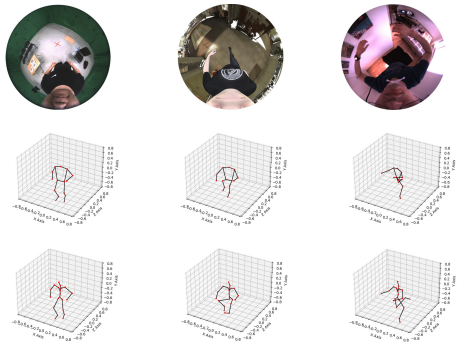


Figure 5. A sample of SMPL pose fitting results for each dataset. The top row displays the original fisheye images, the middle row shows the simplified pose estimation produced by SceneEgo, and the bottom row presents the fitted SMPL skeleton. From left to right: SceneEgo, EgoGTA, and Self-Recorded.

Texture Warping and Transfer

The results of the texture warping and transfer steps demonstrate a substantial deviation from the ground truth texture data provided by the source image. The aligned silhouette renders of the SMPL mesh seldom seem match the silhouette of the subject to a sufficient degree to which a realistic mapping can be produced. This deviation becomes more pronounced as the predicted pose diverges further from the ground truth pose. Furthermore, some deviation between the mesh and the image is also caused by the camera assumptions used to generate the mesh render, as the extrinsic parameters do not always precisely match the position from which the ground truth image was captured. This issue is particularly prominent in the Self-Recorded dataset.

In addition, the segmented images produced by SceneEgo exhibit a moderate margin of error in pixel classification. This leads to the unpredictable appearance of artifacts from the surrounding environment in various areas of the segmented fisheye image, which, depending on their size and position, can significantly impact the warping step of the procedure. This can be clearly seen in figure 6, where parts of the surrounding produce to a highly distorted mapping to the silhouette.

As a result, the mappings produced by the naive warping procedure can be highly inaccurate, as most relational information is lost during the transfer to the mesh, resulting in an unintelligible texture on the mesh's UV map (See Figure 8).

Texture Generation

The texture inpainting step is generally capable of producing realistic and consistent UV maps. When the partially

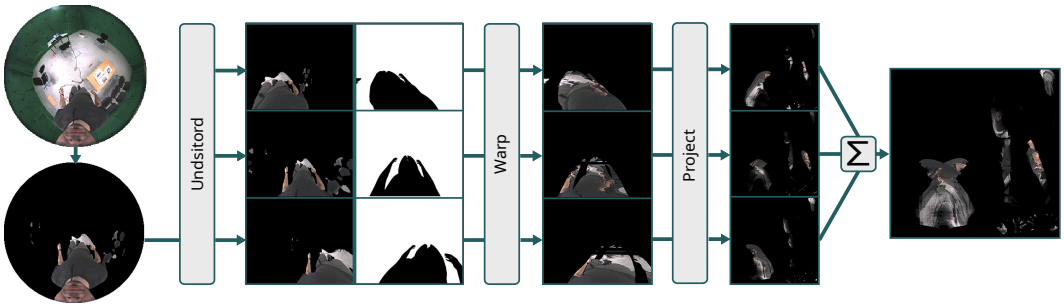


Figure 6. Visualization of the texture warping and mapping process. The undistorted images are first mapped onto the pixels of their respective silhouette before being projected onto the mesh from a viewpoint that produces the same silhouette. Finally, all projected UV maps are combined into a single UV map.

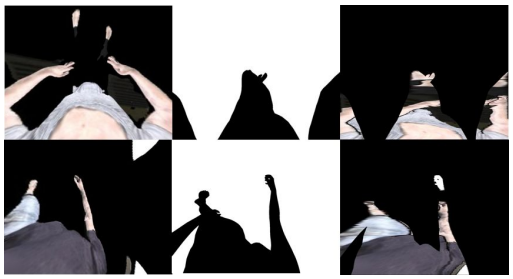


Figure 7. Examples of extreme distortion resulting from perspective misalignment and artifacts in the segmented image. The left image displays the undistorted source image, the middle image shows the silhouette from the same perspective, and the right image presents the texture mapping onto the silhouette.



Figure 8. Inpainting results for the partial UV-Maps of each dataset. The left images show the projected pixel from the warping procedure while the right shows the fully inpainted UV-Map by the SMPLitex generative model. From top to bottom: SceneEgo, EgoGTA and Self-Recorded.

mapped UV map is sufficiently interpretable, the model correctly identifies the color of clothing and extends the texture around the entire circumference of the body. However, the lower information density in certain regions, particularly the lower body, often results in insufficient data, causing the generative model to hallucinate texture information rather than generating it based on the available input. Some examples are Figure 4 bottom left and Figure 4 bottom right.

Hallucination of data becomes more pronounced as the partial mapped UV texture becomes increasingly warped and incomprehensible, leaving the model unable to rely on any meaningful information from the texture. In such cases, however, the generated data tends to approximate the general color scheme of the mapped texture (See Figure 2 bottom right and Figure 3 bottom left).

Discussion

Results Analysis

The results show a generally suboptimal performance across the various datasets. These discrepancies primarily stem from multiple weaknesses in the pipeline, primarily in pose prediction, and the warping process.

The first issue arises from the model's high reliance on accurate pose predictions produced by the SceneEgo model. While the SceneEgo model's joint predictions appear accurate when rendered using the simplified skeleton, their accuracy has proven insufficient when transferred onto the SMPL mesh and used as the basis for texture mapping. An initially inaccurate pose estimation is further amplified by subsequent steps in the pipeline, reducing mapping accuracy in each step.

Moreover, as previously mentioned, the predictions are not constrained by body proportions, leading to poses with asymmetric limb lengths. The limb proportions and asymmetry carry over to the next step, where the simplified joint predictions are fitted to an SMPL skeleton. Since body parameters are not adjusted due to insufficient information and the skeleton's limb lengths are fixed, the fitted pose

further deviates from the predicted joints, compromising into a local minimum in the loss function.

This results in an SMPL mesh that deviates slightly for poses where the arms are closer to the body and more significantly when the arms extend outward. The issue becomes more pronounced when the mesh is rendered from the camera's perspective and alignment is attempted. The naive warping step yields good results for minor deviations from the original pose, particularly when the pose and texture align along the image's height. However, it fails to compute a realistic mapping when the discrepancy between the mesh and texture is too large (see Figure 7).

Another challenge in SceneEgo's segmentation process is its misclassification of background elements as part of the subject, significantly degrading the quality of the warped texture by incorporating background artifacts (See Figure 7). This issue highlights the limitations of the naive warping approach, which fails to account for body part differences in lower-accuracy cases. The results suggest that a more complex warping procedure incorporating semantic context of body part segments is required, exceeding the capabilities of the method used here. Furthermore, while the segmented image proves sufficient for internal computations within SceneEgo's pose prediction pipeline, it lacks the accuracy necessary for the texture mapping task applied in this method. However, when the texture closely aligns with the mesh, the method can still yield reasonable results, though the conditions where these cases apply are generally the exception rather than the norm.

Finally, the resulting mesh is constrained by the SMPLitex generative model, which in some cases deviates from the partial UV data. While the generated data for clothing generally aligns with the overall color scheme of the prior, the model occasionally exhibits significant deviations from the input, creating sharp transitions. For example, this is visible in Figure 1, where the model is given one skin color but transitions to a darker skin tone for the rest of the body. However, unlike other issues observed in this method, this deviation could be mitigated entirely through adjustments in the preceding steps.

Possible Improvements

One potential improvement is adjusting body parameters of the SMPL mesh to minimize the warping needed when mapping the texture to the silhouette. In theory, the texture, combined with camera parameters, provides sufficient information to infer visible body dimensions. However, for this adjustment to be effective, a more precise alignment between the texture and the predicted mesh silhouette would be a necessary prerequisite.

Furthermore, to compensate for inaccurate estimations, a more sophisticated approach to warping the texture onto the predicted pose's mesh could be employed. One potential improvement involves segmenting the target texture into semantic body parts. These segments could, for example, be inferred directly from the joint predictions, provided they align with the 2D pose. Once segmented, the texture regions can be mapped to their corresponding areas on the mesh's silhouette, which would be pre-labeled accordingly prior to projection.

Alternative approaches

Before implementing a mesh-based approach, a Gaussian-based procedure was explored. However, this approach proved highly restrictive in the early stages of development. Generating a point cloud from undistorted fisheye images was particularly challenging, as the body remained visible from only a single perspective across multiple frames, making depth estimation from a single perspective impossible.

In later development, the Sceneego model was used to generate a synthetic depth map from the fisheye image. While this depth map could initialize Gaussians in 3D space, it would produce a point cloud limited to the depth map's resolution, with no direct relation to the predicted skeleton and its joints. Integrating both would require warping point locations onto the SMPL mesh surface in the predicted pose. However, this would further complicate the warping process, as additional depth alignment would also need to be considered.

Initially, an alternative approach was considered for generating the visual representation based on the SMPL skeleton fitted to the texture. Neural body models, which learn a 3D representation from 2D inputs captured from different perspectives—such as DANBO [9] for implicit NeRF representations or HuGS [6] for explicit Gaussian rendering, were explored as potential solutions. However, texture misalignment, the limited number of available perspectives, and the unavailability of source code for HuGS made utilizing either model impractical.

Limitations and future work

Beyond alignment challenges, a fundamental limitation of single-perspective reconstruction is the lack of texture information for body parts that are barely or not visible in the fisheye view. While areas, like the face and upper body, may be transferred depending on the camera's position in relation to the body, regions such as the back must

cannot be inferred from a singular view. This makes a fully realistic reconstruction impossible without additional viewpoints

Additionally, this procedure does not model joint rotation and instead relies a simplified assumption by always mapping textures to the anterior of arms and legs. To address this, the aforementioned method proposed in [11] could be applied. Since it also models finger positions, this additional context could help infer arm rotation and improve the accuracy of texture mapping.

Future work could integrate a more advanced pose prediction model alongside a more sophisticated warping procedure for finer adjustments, enabling more realistic results and providing better context for the generative process. Improved alignment between pose and texture could also facilitate the use of a Gaussian splatting approach, serving as a conditional input for Gaussian generative models such as [3].

Conclusion

In conclusion, this method demonstrates that pose and visual reconstruction from a single egocentric fisheye frame is feasible but requires high accuracy at each stage of the pipeline to achieve realistic results. Even small errors tend to propagate through the process, significantly affecting the final output. Future research could focus on developing or integrating a more precise pose prediction method with improved segmentation quality, as well as refining the warping procedure. These improvements could serve as a solid foundation for various approaches, whether based on rendered meshes or Gaussian representations.

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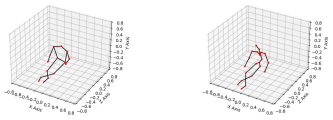
Appendix

Source Code

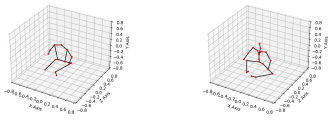
The source code for this procedure can be found Here.
Note that Model weights, as well as SceneEgo and SMPLitex are not included in the project files and have to be installed separately.

Supplementary Figures

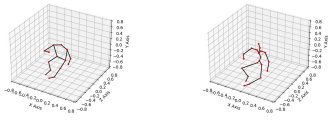
This section contains supplementary figures and graphs for the sample data described in the main results section.



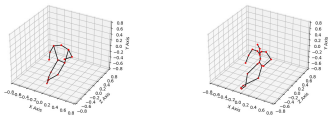
(a) Top left



(b) Top right

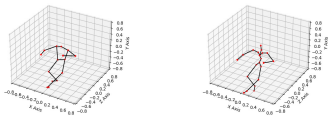


(c) Bottom Left

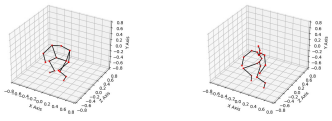


(d) Bottom Right

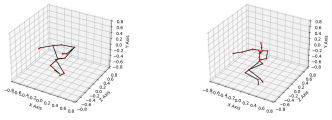
Figure 9. Skeleton fitting results for each Self-Recorded sample in figure 4.



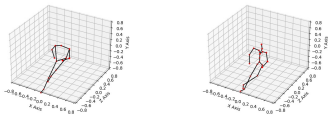
(a) Top Left



(b) Top Right

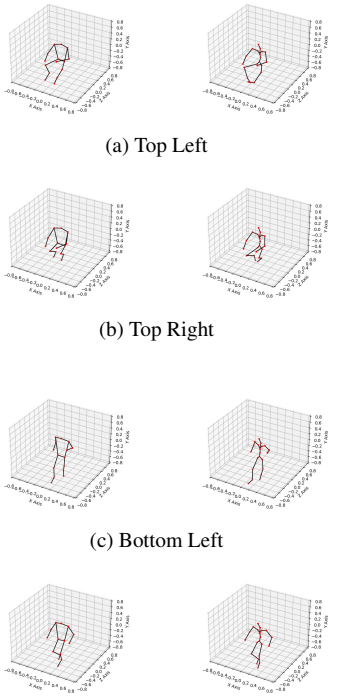


(c) Bottom Left



(d) Bottom Right

Figure 10. Skeleton fitting results for each EgoGTA sample in figure 3.



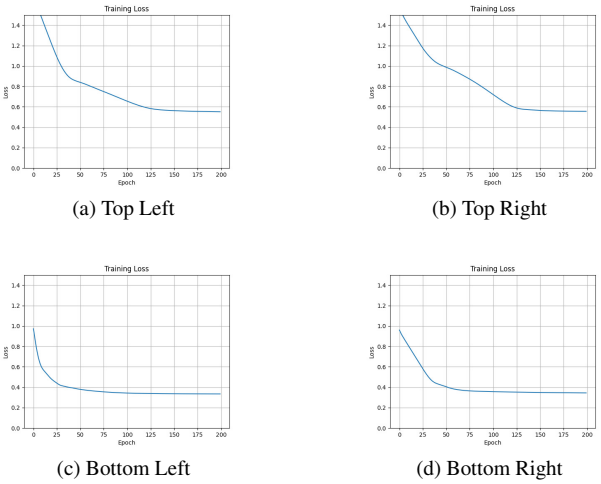
(a) Top Left

(b) Top Right

(c) Bottom Left

(d) Bottom Right

Figure 11. Skeleton fitting results for each SceneEgo sample in figure 2.



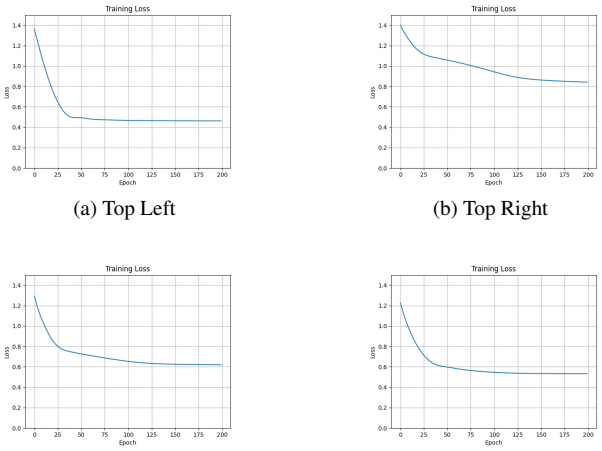
(a) Top Left

(b) Top Right

(c) Bottom Left

(d) Bottom Right

Figure 13. Plot of the loss at each iteration of the fitting procedure over 200 iterations for each SceneEgo sample shown in Figure 2.



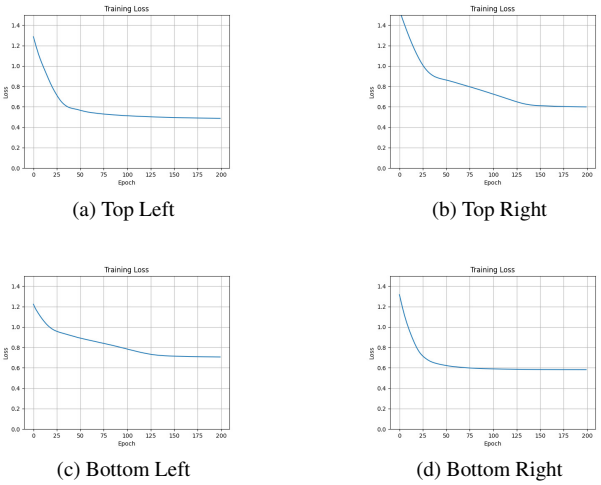
(a) Top Left

(b) Top Right

(c) Bottom Left

(d) Bottom Right

Figure 12. Plot of the loss at each iteration of the fitting procedure over 200 iterations for each self-recorded sample shown in Figure 4.



(a) Top Left

(b) Top Right

(c) Bottom Left

(d) Bottom Right

Figure 14. Plot of the loss at each iteration of the fitting procedure over 200 iterations for each EgoGTA sample shown in Figure 3.